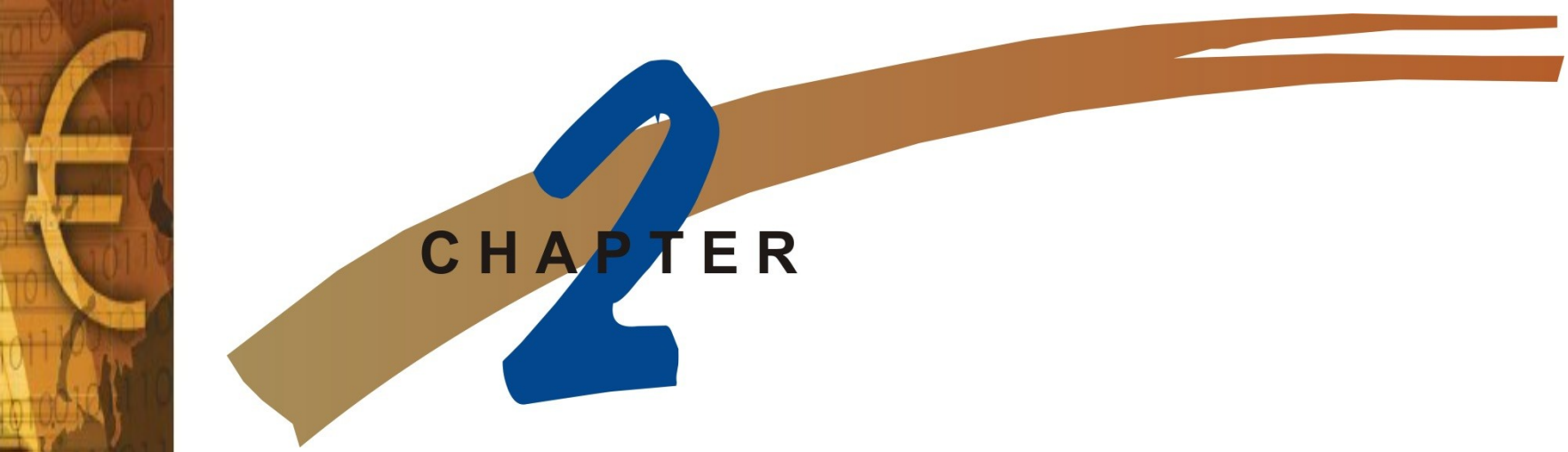


The World Wide Web

ELECTRONIC COMMERCE
From Vision to Fulfillment

Third Edition

Elias M. Awad

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The focus of this chapter is on several learning objectives

- The rising tide of the Internet
- The makeup of the Web
- The main Web search elements
- The main elements of Web research and Web research tips
- The search process and some important facts to remember about Web research
- How to optimize Web sites
- The role of the ISP
- Web fundamentals including the makeup of the URL
- Internet services

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The Beginnings of the Internet

- Internet is the infrastructure that links thousands of networks together
- Internet owes its existence to the Pentagon and the cold war
  - Original networked sites were military installations, universities, and business firms with defense department contracts
  - Initial goal was to design a network that would maintain the safe transition of data between military computers at select sites through redundant communication routes

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The Beginnings of the Internet *(cont'd)*

- Researchers devised a way of bundling information into packets that carried the network address of the recipient
  - Each packet is sent into a so-called network "cloud" across the vast array of computers on the network
  - Each computer checks to see if the information belongs to any of its clients and forwards it to the next computer to which it might belong
  - Once claimed by the right computer, the packet is opened to reveal the message
  - This message delivery system is moved by a protocol

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The Beginnings of the Internet *(cont'd)*

- At first, Internet traffic was government related and government subsidized: No ordinary person or company could use the Internet
  - April 1995 the U.S. government relinquished control of the Internet to independent governing bodies, which relaxed entry for almost everyone

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The Beginnings of the Internet *(cont'd)*

- The Internet today offers a variety of services including:
  - E-mail
  - File transfer
  - Interest group membership
  - Multimedia displays
  - Real-time broadcasting
  - Shopping opportunities
  - Access to remote computers
  - Quick and easy transmission of information among computers worldwide

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The Making of the World Wide Web

- Tim Berners-Lee wrote a program that allowed information highlighted in a document to link to other documents on a computer network with a mouse click
- Hypertext is text that contains keywords to connect to other documents
- Keywords called links, also referred to as hyperlink, connects current document to another location in the same document or to another document on the same host computer
- Hypertext for the Web can be generated with a language called HyperText Markup Language (HTML)

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The Making of the World Wide Web *(cont'd)*

- **The World Wide Web *(also known as the Web)* is the universe of network-accessible information and the embodiment of the human knowledge**
  - **A cluster of software, protocols, and standards**
  - **An organization of files designed around a group of Internet servers programmed to handle requests from browser software that resides on users' PC**
  - **Single document can be perceived to stretch - weblike - throughout the world (See www.w3.org/www)**
- **When a document is accessed the components are pulled from different computer worldwide and integrated in the document displayed on the user's screen**

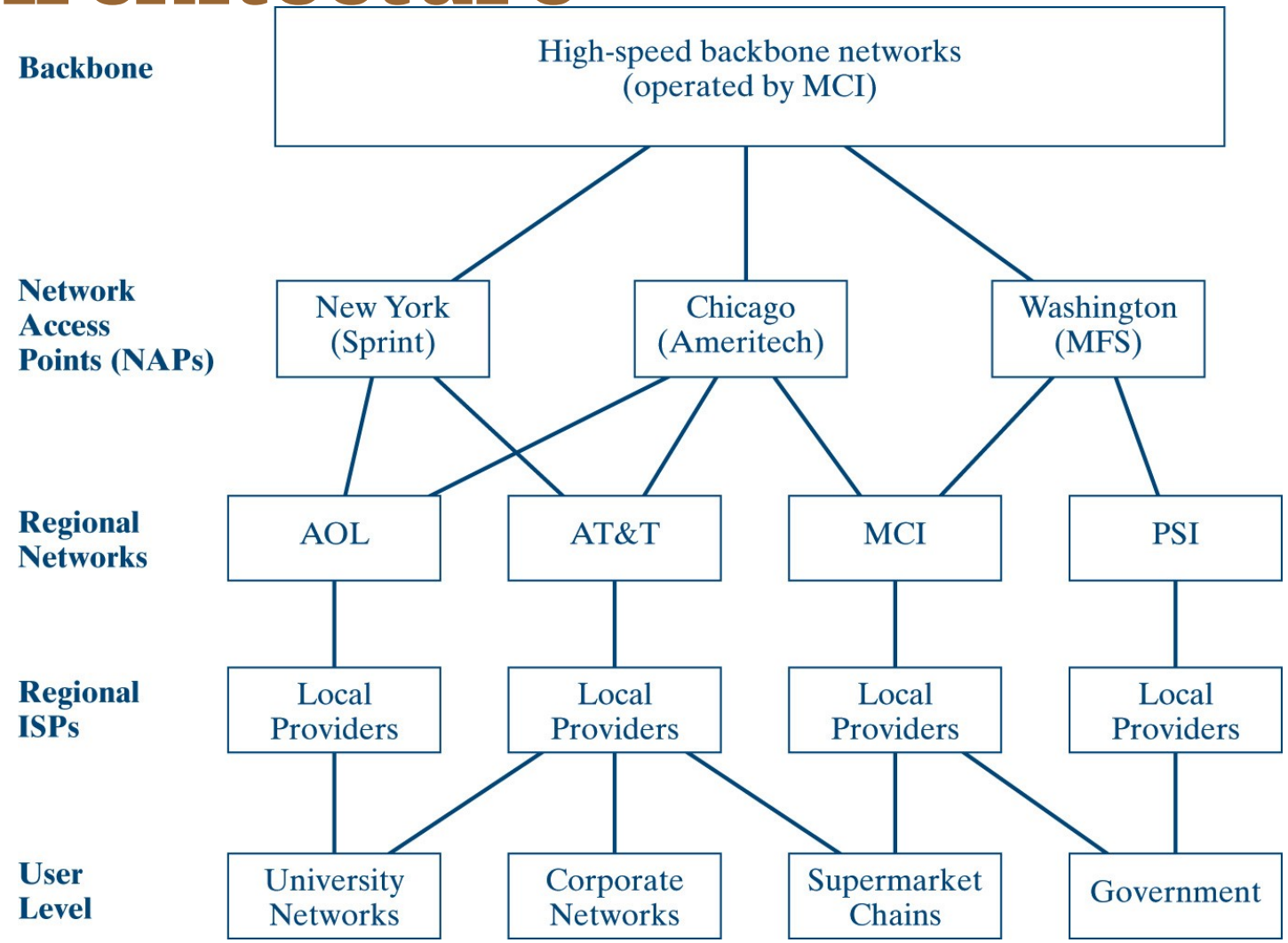
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Physical Structure of the Internet

- Physical structure, or architecture, of the Internet is hierarchical
  - High-speed backbones are at the top
  - Bulk of Internet traffic is fed onto the backbone via network access points (NAPs)
  - Regional and individual networks at the bottom
- The Internet uses a common set of communication protocols called the TCP/IP (transmission control protocol / Internet protocol) suite that provide the basis for operating the Internet

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General Internet Network Architecture



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Key Protocols Accessible on the Web

- E-mail **- the protocol for e-mail is Simple Mail Transport Protocol, or SMTP**
- **HTTP - Hypertext Transfer Protocol makes possible transmission of hypertext over networks**
- **VoIP - Voice over Internet Protocol makes it possible to place a telephone call over the Web**

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Web Search Elements

- Internet contains thousands of Web sites dedicated to tens of thousands of topics
- Key elements that make the search process feasible:
  - Browser
  - Plug-Ins
  - Search Engine

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The Browser *(cont'd)*

- A browser is a piece of software that allows users to navigate the Web
  - Netscape Navigator
  - Microsoft Internet Explorer
  - Firefox
  - Opera

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The Browser *(cont'd)*

- **A browser is a Web client program that uses Hypertext Transfer Protocol (HTTP) to make requests of Web servers throughout the Internet on behalf of the browser user**
  - ***Text-only mode* such as Lynx**
  - ***Graphic mode* involves a graphical software program that retrieves text, audio, and video**

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Plug-Ins

- Software programs configured to a Web browser to improve its capabilities
- Working together with plug-ins, browsers today offer seamless multimedia experiences
- A popular plug-in on the Web is Adobe Shockwave flash
- Microsoft developed software called Active X, which makes plug-ins unnecessary
  - This software makes it possible to embed animated objects and data on Web pages
  - Being a Microsoft product, Active X works best with Microsoft's Internet Explorer browser

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Multimedia

- Since 1999, it is now common to use the Web to listen to audio and watch video - prerecorded or live off the Internet
- Streaming media is audio or video that begin to play as it downloads (streaming), done through buffering
  - Buffering is used to minimize the wait time between downloading and actual viewing of the material
- RealPlayer and Windows Media Player are alternative options for the broadcast of real-time (live) events
- Shockwave is multimedia software that allows for an entire multimedia display of audio, graphics, animation, and sound
- Live Cam software essentially is a video camera that digitizes images and transmits them in real time to a Web server
- Chat programs make it convenient for people to “talk” to each other in real time by typing messages and receiving responses (i.e. America Online’s Instant Messenger)

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The Search Engine

A search process begins with a search engine:

a Web site or a database, along with the tools to generate that database and search its contents for "keywords" that describe what you're looking for.

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Other Definitions for Search Engines

- A software program that collects and indexes Internet resources and provides a keyword search system allowing the user to identify and retrieve resources based on words, phrases, or patterns within those documents
- A Web-based system for searching the information available on the Web
- An automated system that relies on a software agent (otherwise known as spiders, robots or crawlers) that explores the World Wide Web following links from site to site and catalogs relevant text and content, storing Web pages and creating a customized index based on the user's query of the search engine's database

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More on Search Engines

- Two main elements of Web research are indexes and search engines
- An index can help a searcher acquire general information or gain a feel for the general topic
- An index can be hierarchical or alphabetical
  - Hierarchical indexing leads from general to specific topics
  - Alphabetical indexing contains sources that focus on a specific topic or area of concern

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Other Components of a Web Search Engine

- A spider is a program that roams the Web from link to link, identifying and scanning pages
- A spider is software unique to a search engine that allows users to query the index and returns results in relevancy-ranked order (alphabetical)
- Search Engine Improvements:
  - First-generation search engine returns results in schematic order, constructing a term relevancy rating of each hit and presenting search results in this order, also called “on the page” ranking.
  - Second-generation search engine organizes search results by peer ranking concept, domain, or site rather than by relevancy, also called “off the page” information.

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Search Facts to Remember

- People look up Web sites with search engines.
- People usually use bookmarks to visit their favorite Web sites.
- A Web site must be quick and current.
- A Web site should address the privacy and navigation concerns of the user.
- The “bottleneck” problem
- People are reluctant to pay to surf a Web site.

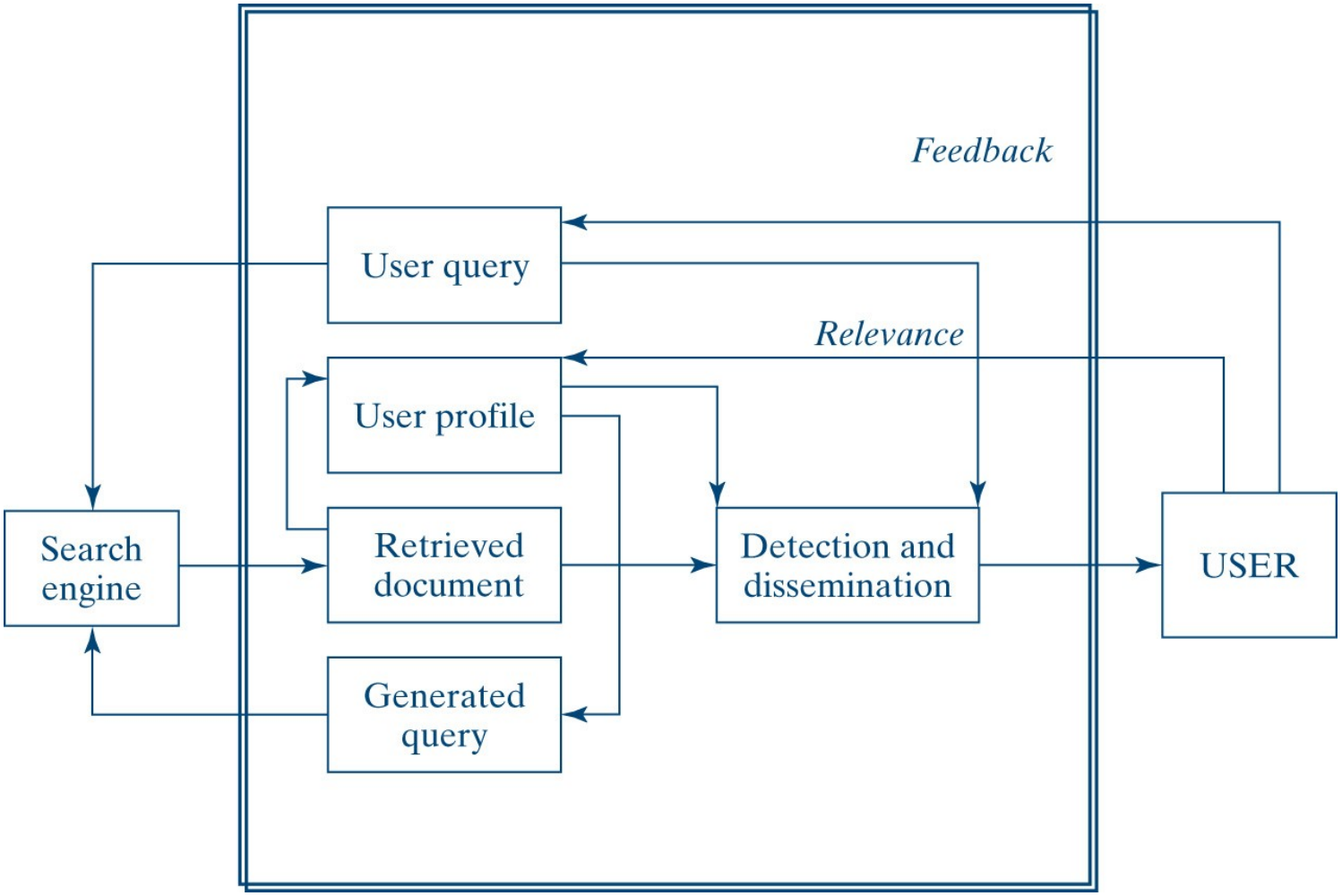
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Search Engines Are Getting Smarter

- Experts are working feverishly at making search engines more intelligent
- An example of incorporating “intelligence” into search engines is a software agent called Query Tracker that supplements a user’s query with its own, and it gains in performance with prolonged use and feedback.
- Another intelligent search engine application is IBM’s WebFountain that determines whether an entity is a person’s name, a corporate logo, a product, or a discount and then goes ahead and attaches a metadata tag to it.
- On the drawing board is “thinking in pictures” as an alternative way to search the Web

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Query Tracker



Source: Adapted from Anthes, Gary H., "Search For Tomorrow" Computerworld, April 5, 2004, 26.

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Search Engine Optimization

- A way of trying to increase the number of visitors to a Web site by ranking high in the search results displayed by a search engine
- One way to optimize is via hyperlinks

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Tips for Search Engine Optimization

- General keywords are nowhere as good as specific keyword phrases
- Check the Web site of the competition for ideas
- Think of what visitors would search for in the page you're optimizing
- Include the most important keyword phrases in heading tags on your page
- Finalize the list of keyword phrases for the pages you optimize
- The title tag of your page is the most important factor to consider
- Visitors as well as search engines read your pages by looking at keywords to see what you have to offer
- Many people have a false impression that good metatags are all that is needed to achieve good listings in the search engine
- There are two metatags that can help in listing your Web site: meta keywords and meta descriptions

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Internet Service Providers

- Internet service provider (ISP) is a company that links users to the Internet for a fee
- Services offered by ISPs
  - Linking consumers and businesses to the Internet
  - Monitoring and maintaining customers' Web sites
  - Providing network management and system integration
  - Providing backbone access services for other ISPs (like PSI and UUNET)
  - Offering payment systems for online purchases

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Internet Service Providers (cont'd)

- Initially, the cost for Internet access often exceeded $1,000 per month
- Many of today's ISPs offer unlimited access for as low as $5 per month
- Many local governments are funding the use of the Internet because of its political, educational, and commercial benefits
- The problem for some ISPs is sudden growth without advance planning to accommodate that growth
- A well managed ISP requires:
  - Professional management
  - A highly skilled technical staff
  - Healthy budget to bring the technology in line with the voracious
  - Ensure a balance between creativity and control and between managing growth and a stable technical infrastructure appetite of today's consumer

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Stability and Reliability of the Web

- No one single agency or company owns the Internet
- Each company on the Internet owns its own network
- Links between these companies and the Internet are owned by telephone companies and ISPs
- The organization that coordinates Internet functions is the Internet Society
- Internet is designed to be indefinitely extendable
- Reliability depends primarily on the quality of service providers' equipment

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Web Fundamentals

- The WWW is a global network of millions of Web servers and Web browsers connected by the hypertext transfer protocol (HTTP)
- The WWW is a giant client/server system
  - Content is held by Web servers and requested by clients or browsers
  - Clients display the information sent by the Web server on their monitors
  - Web servers provide pages of multimedia information in seconds
  - The most important element of a Web site is its links to other pages within the site or across sites

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Web Fundamentals - URLs and HTTP

- Uniform Resource Locators (URLs) are central to the Web in e-commerce
- A URL such as http://www.virginia.edu consists of two key parts:
  - http:// (Hypertext Transport Protocol) is a protocol designator
  - www.virginia.edu is the server name:
    - *www* after the double slash tells the network that the material requested is located on a dedicated Web server
    - *Virginia* is the name of the Web site requested
    - *Edu* is a code for the domain of the Web server indicating that the site is an *edu*cational institution

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Security Protocols

- Two main security protocols
  - Secure Sockets Layer (SSL) is a protocol for transmitting private information in a secure way over the Internet
  - Secure HTTP (S-HTTP) provides various security features such as client/server authentication and allows Web clients and servers to specify privacy capabilities
- Security is a major concern

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Key Terms for Internet Literacy

- Internet service provider (ISP) is a company that links users to the Internet for a fee; the entrance ramp to the Internet.
- A Browser is a software program loaded on a PC that allows you to access or read information stored on the Internet. It is the client program that enables you to interface with the Internet.
- Server is the destination point on the Internet where the information you are seeking is stored.
- Electronic mail (e-mail) is probably the most popular and abused network application across all user categories.

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Key Terms for Internet Literacy (*cont'd*)

- File transfer protocol (FTP) is a standard protocol that allows you to copy files from computer to computer
- Telnet is a basic Internet protocol that allows you to log on to a computer and access files from a remote location as if they were local files
- Bulletin board systems (BBS) is a computer-based meeting and announcement system that allows local people to exchange information free of charge. A BBS generally has a simple interface to the Internet for users to access services like e-mail and NetNews.

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Key Terms for Internet Literacy (cont'd)

- Web pages are written in a language called Hypertext Markup Language (HTML)
  - Language specifies display features that visually structure a page
  - Best known for publishing static information (flows in one direction)
- How to allow the user to interact with a Web site?
  - Common gateway interface (CGI) scripts execute a process on the server. When executed, the script passes data (posted form) provided by the customer for verification and action.
  - Java programming language or Java makes it possible for the customer to interact directly with the program on the screen.

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Managerial Implications

- The Web has changed the way business and information technology work together.
- E-commerce is transforming the Internet from a “browse-and-surf” environment into a mammoth information exchange.
- The important thing is to keep an eye on the technologies, as they evolve, and to be familiar with the changes before taking a dive into the Internet.
- Strategize first, test the waters, and be sure you have a unique product supported by qualified staff to follow up on the Web traffic that it attracts.

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Chapter Summary

- The Web is the fastest growing, most user-friendly, and most commercially popular technology to date.
- The Internet owes its existence to the Pentagon, where it originally was created for military research.
- The Internet is physically hierarchical. High-speed backbones are at the top, with regional and individual networks at the bottom.
- Internet service providers link commercial traffic to its destination.

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Chapter Summary (cont'd)

- The Internet has many uses; it also has many limitations.
- World Wide Web is a global hypertext network of millions of Web servers and browsers connected by hypertext transfer protocol (HTTP) and its many derivatives.
- It is important to learn the language of the Internet before starting an e-commerce project.
- The Internet and the Web have changed the way business and technology work together.

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